Sheet #1 (Data type - Conditions)

A. Say Hello With C++

1 second, 256 megabytes

Given a name S. Print "Hello, (name)" without parentheses.

Input
Only one line containing a string S.

Output
Print "Hello, " without quotes, then print name.

input
```
programmer
```
output
```
Hello, programmer
```

B. Basic Data Types

1 second, 256 megabytes

The following lines show some C++ data types, their format specifiers and their most common bit widths:

- **int** : 32 Bit integer.
- **long long** : 64 bit integer
- **Char** : 8 bit Characters & symbols
- **Float** : 32 bit real value
- **Double** : 64 bit real value

Reading

To read a data type, use the following syntax:

```
cin >> VariableName;
```

For example, to read a character followed by a double:

```
char ch;
double d;
cin >> ch >> d;
```

Printing

To print a data type, use the following syntax:

```
cout << VariableName;
```

For example, to print a character followed by a double:

```
char ch = 'd';
double d = 234.432;
cout << ch << " "<< d;
```

Input
Only one line containing the following space-separated values: **int, long long, char, float** and **double** respectively.

Output
Print each element on a **new line** in the same order it was received as input.

Don't print any extra spaces.

input
```
3 12345678912345 a 334.23 14049.30493
```
output
```
3
12345678912345
a
334.23
14049.3
```

C. Simple Calculator

1 second, 256 megabytes

Given two numbers X and Y. Print the **summation** and **multiplication** and **subtraction** of these **2** numbers.

Input
Only one line containing two separated numbers X, Y ($1 \le X, Y \le 10^5$).

Output
Print **3** lines that contain the following in the same order:

1. "X + Y = **summation** result" without quotes.
2. "X * Y = **multiplication** result" without quotes.
3. "X - Y = **subtraction** result" without quotes.

input
```
5 10
```
output
```
5 + 10 = 15
5 * 10 = 50
5 - 10 = -5
```

Be careful with spaces.

D. Difference

1 second, 256 megabytes

Given four numbers A, B, C and D. Print the result of the following equation :

$X = (A * B) - (C * D)$.

Input
Only one line containing 4 separated numbers A, B, C and D ($-10^5 \le A, B, C, D \le 10^5$).

Output
Print "Difference = " without quotes followed by the equation result.

input
```
1 2 3 4
```
output
```
Difference = -10
```

input
```
2 3 4 5
```
output
```
Difference = -14
```

input
```
4 5 2 3
```
output
```
Difference = 14
```

E. Area of a Circle

1 second, 256 megabytes

Given a number R calculate the **area** of a circle using the following formula:

Area = $\pi * R^2$.

Note: consider π = **3.141592653**.

Input

Only one line containing the number R ($1 \le R \le 100$).

Output

Print the calculated **area**, with **9** digits after the decimal point.

input
2.00
output
12.566370612

* Use the data type double for this problem.

** Use setprecision(9) to print 9 digits after decimal point.

*** you can use function **setprecision** that are in **#include<iomanip>** library for Example :

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
     cout << fixed << setprecision(9);
     // your code.
}
```

F. Digits Summation

0.25 seconds, 64 megabytes

Given two numbers N and M. Print the **summation** of their **last digits**.

Input

Only one line containing two numbers N, M $(0 \le N, M \le 10^{18})$.

Output

Print the answer of the problem.

input
13 12
output
5

First Example :

last digit in the first number is **3** and **last digit** in the second number is **2**.

So the answer is: (**3 + 2 = 5**)

G. Summation from 1 to N

0.25 seconds, 256 megabytes

Given a number N. Print the **summation** of the numbers that is between **1** and N **(inclusive)**.

$$\sum_{i=1}^{N} i$$

Input

Only one line containing a number N $(1 \le N \le 10^9)$

Output

Print the **summation** of the numbers that are between **1** and N **(inclusive)**.

input
3
output
6

input
10
output
55

First Example :

the numbers between 1 and 3 are **1,2,3** .

So the answer is: (**1 + 2 + 3 = 6**)

Second Example :

the numbers between 1 and 10 are **1,2,3,4,5,6,7,8,9,10**.

So the answer is: (**1 + 2 + 3 + 4 + 5 + 6 +7 +8 + 9 + 10 = 55**)

H. Two numbers

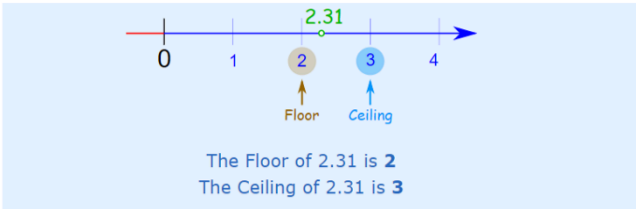
1 second, 256 megabytes

Given **2** numbers A and B. Print **floor**, **ceil** and **round** of A/B

Note:

- **Floor**: Is a mathematical function that takes a real number X and its output is the **greatest** integer **less than** or **equal to** X.
- **Ceil**: Is a mathematical function that takes a real number X and its output is the **smallest** integer **larger than** or **equal to** X.
- **Round**: Is a mathematical function that takes a real number X and its output is the **closest** integer to that number X.

The round of 7.3 is **7**
The round of 7.5 is **8**
The round of 7.7 is **8**



The Floor of 2.31 is **2**
The Ceiling of 2.31 is **3**

For more clarification visit the links in the notes below.

Input

Only one line containing two numbers A and B $(1 \le A, B \le 10^3)$

Output

Print **3** lines that contain the following in the same order:

1. "floor A / B = **Floor result**" without quotes.
2. "ceil A / B = **Ceil result**" without quotes.
3. "round A / B = **Round result**" without quotes.

input

```
10 3
```

output

```
floor 10 / 3 = 3
ceil 10 / 3 = 4
round 10 / 3 = 3
```

input

```
10 4
```

output

```
floor 10 / 4 = 2
ceil 10 / 4 = 3
round 10 / 4 = 3
```

input

```
10 6
```

output

```
floor 10 / 6 = 1
ceil 10 / 6 = 2
round 10 / 6 = 2
```

Links:

- For Rounding method visit: https://www.mathsisfun.com/numbers/rounding-methods.html.
- For Flooring and Ceiling method visit: https://www.mathsisfun.com/sets/function-floor-ceiling.html.

I. Welcome for you with Conditions

1 second, 64 megabytes

Given two numbers A and B. Print "**Yes**" if A is **greater than or equal to** B. Otherwise print "**No**".

Input

Only one line containing two numbers A and B ($0 \le A, B \le 100$).

Output

Print "Yes" or "No" according to the statement.

input

```
10 9
```

output

```
Yes
```

input

```
5 5
```

output

```
Yes
```

input

```
5 7
```

output

```
No
```

J. Multiples

1 second, 256 megabytes

Given two numbers A and B. Print "Multiples" if A is **multiple** of B or **vice versa**. Otherwise print "No Multiples".

Input

Only one line containing two numbers A, B ($1 \le A, B \le 10^6$)

Output

Print the "Multiples" or "No Multiples" corresponding to the read numbers.

input

```
9 3
```

output

```
Multiples
```

input

```
6 24
```

output

```
Multiples
```

input

```
12 5
```

output

```
No Multiples
```

***A is said to be Multiple of B if $\boldsymbol{B}$ **is divisible by** $\boldsymbol{A}$.

First Example :

9 is divisible by **3** , So the answer is: Multiples.

Second Example :

6 is **not divisible** by **24** but

24 is divisible by **6** , So the answer is: Multiples.

Third Example :

12 is not divisible by **5** and **5** is not divisible by **12**.

So the answer is: No Multiples.

K. Max and Min

0.25 seconds, 64 megabytes

Given 3 numbers A, B and C, Print the **minimum** and the **maximum** numbers.

Input

Only one line containing 3 numbers A, B and C ($-10^5 \le A, B, C \le 10^5$)

Output

Print the **minimum** number followed by a single space then print the **maximum** number.

input

```
1 2 3
```

output

```
1 3
```

input

```
-1 -2 -3
```

output

```
-3 -1
```

input

```
10 20 -5
```

output

```
-5 20
```

L. The Brothers

1 second, 256 megabytes

Given two person names.

Each person has {"the first name" + "the second name"}

Determine whether they are brothers or not.

Note: The two persons are brothers if they **share the same second name**.

Input

First line will contain two Strings F_1, S_1 which donates the first and second name of the 1st person.

Second line will contain two Strings F_2, S_2 which donates the first and second name of the 2nd person.

Output

Print **"ARE Brothers"** if they are brothers otherwise print **"NOT"**.

input

```
bassam ramadan
ahmed ramadan
```

output

```
ARE Brothers
```

input

```
ali salah
ayman salah
```

output

```
ARE Brothers
```

input

```
ali kamel
ali salah
```

output

```
NOT
```

M. Capital or Small or Digit

1 second, 256 megabytes

Given a letter X. Determine whether X is Digit or Alphabet and if it is Alphabet determine if it is **Capital Case** or **Small Case**.

Note:

- Digits in ASCII '0' = 48,'1' = 49 ....etc
- Capital letters in ASCII 'A' = 65, 'B' = 66 ....etc
- Small letters in ASCII 'a' = 97,'b' = 98 ....etc

Input

Only one line containing a character X which will be a capital or small letter or digit.

Output

Print a single line contains "**IS DIGIT**" if X is **digit** otherwise, print "**ALPHA**" in the first line followed by a new line that contains "**IS CAPITAL**" if X is a **capital** letter and "**IS SMALL**" if X is a **small letter**.

input

```
A
```

output

```
ALPHA
IS CAPITAL
```

input

```
9
```

output

```
IS DIGIT
```

input

```
a
```

output

```
ALPHA
IS SMALL
```

** recommended to read this to know more about ASCII Code https://www.javatpoint.com/ascii.

N. Char

0.25 seconds, 64 megabytes

Given a letter X. If the letter is **lowercase** print the letter after converting it from **lowercase letter to uppercase letter**. Otherwise print the letter after converting it from **uppercase letter to lowercase letter**

Note : **difference between 'a' and 'A' in ASCII is 32** .

Input

Only one line containing a character X which will be a **capital** or **small** letter.

Output

Print the answer to this problem.

input

```
a
```

output

```
A
```

input

```
A
```

output

```
a
```

O. Calculator

1 second, 256 megabytes

Given a mathematical expression. The expression will be one of the following expressions: $A + B$, $A - B$, $A * B$ and A/B.

Print the **result** of the mathematical expression.

Input

Only one line contains A, S and B $(1 \le A, B \le 10^4)$, S is either $(+, -, *, /)$.

Output

Print the **result** of the mathematical expression.

input

```
7+54
```

output

```
61
```

input

```
17*10
```

output

```
170
```

For the dividing operation you should print the division without any fractions.

P. First digit !

0.25 seconds, 64 megabytes

Given a number X. Print "EVEN" if the first digit of X is **even number**. Otherwise print "ODD".

For example: In **4569** the first digit is **4**, the second digit is **5**, the third digit is **6** and the fourth digit is **9**.

Input
Only one line containing a number X $(999 < X \leq 9999)$

Output
If the first digit is even print "$EVEN$" otherwise print "ODD".

input
```
4569
```
output
```
EVEN
```

input
```
3569
```
output
```
ODD
```

Second Example :

In **3569** the first digit is **3** and its ODD.

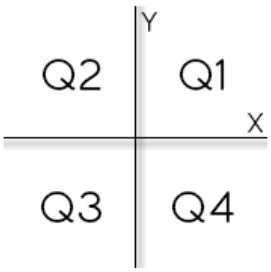
Q. Coordinates of a Point

1 second, 256 megabytes

Given two numbers X, Y which donate coordinates of a point in 2D plan. Determine in which quarter does it belong.

Note:

- Print **Q1, Q2, Q3, Q4** according to the quarter in which the point belongs to.
- Print **"Origem"** If the point is at the origin.
- Print **"Eixo X"** If the point is over X axis.
- Print **"Eixo Y"** if the point is over Y axis.



Input
Only one line containing two numbers X, Y $(-1000 \leq X, Y \leq 1000)$.

Output
Print the answer to problem above.

input
```
4.5 -2.2
```
output
```
Q4
```

input
```
0.1 0.1
```
output
```
Q1
```

R. Age in Days

1 second, 256 megabytes

Given a Number N corresponding to a person's age (in days). Print his age in years, months and days, followed by its respective message "years", "months", "days".

Note: consider the whole year has **365** days and **30** days per month.

Input
Only one line containing a number N $(0 \leq N \leq 10^6)$.

Output
Print the output, like the following examples.

input
```
400
```
output
```
1 years
1 months
5 days
```

input
```
800
```
output
```
2 years
2 months
10 days
```

input
```
30
```
output
```
0 years
1 months
0 days
```

S. Interval

1 second, 256 megabytes

Given a number X. Determine in which of the following intervals the number X belongs to:

[0,25], (25,50], (50,75], (75,100]

Note:

- if X belongs to any of the above intervals print "Interval " followed by the interval.
- if X **does not belong** to any of the above intervals print **"Out of Intervals"**.
- The symbol **'('** represents greater than.
- The symbol **')'** represents smaller than.
- The symbol **'['** represents greater than or equal.
- The symbol **']'** represents smaller than or equal.

For example:

[0,25] indicates numbers between **0** and **25.0000**, including both.

(25,50] indicates numbers greater than **25**: **(25.00001)** up to **50.0000000**.

Input
Only one line containing a number X $(-1000 \leq X \leq 1000)$.

Output
Print the answer to the problem above.

input
```
25.1
```
output
```
Interval (25,50]
```

input

```
25.0
```

output

```
Interval [0,25]
```

input

```
100.0
```

output

```
Interval (75,100]
```

input

```
-25.2
```

output

```
Out of Intervals
```

T. Sort Numbers

0.25 seconds, 256 megabytes

Given three numbers A, B, C. Print these numbers in ascending order followed by a blank line and then the values in the sequence as they were read.

Input

Only one line containing three numbers A, B, C (- $10^6 \le A, B, C \le 10^6$)

Output

Print the values in ascending order followed by a blank line and then the values in the sequence as they were read.

input

```
3 -2 1
```

output

```
-2
1
3

3
-2
1
```

input

```
-2 10 0
```

output

```
-2
0
10

-2
10
0
```

U. Float or int

1 second, 256 megabytes

Given a number N. Determine whether N is **float number** or **integer number**.

Note:

- If N is **float number** then print "**float**" followed by the **integer** part followed by **decimal** part separated by space.
- If N is **integer number** then print "**int**" followed by the **integer** part separated by space.

For more clarification see the examples below.

Input

Only one line containing a number N $(1 \le N \le 10^3)$

Output

Print the answer required above.

input

```
234.000
```

output

```
int 234
```

input

```
534.958
```

output

```
float 534 0.958
```

V. Comparison

1 second, 256 megabytes

Given a comparison symbol S between two numbers A and B. Determine whether it is ***Right*** or ***Wrong***.

The comparison is as follows: A < B, A > B, A = B.

Where A, B are two integer numbers and S refers to the sign between them.

Input

Only one line containing A, S and B respectively (-100 $\le A, B \le$ 100), S can be ('<', '>','=') without the quotes.

Output

Print "Right" if the comparison is true, "Wrong" otherwise.

input

```
5 > 4
```

output

```
Right
```

input

```
9 < 1
```

output

```
Wrong
```

input

```
4 = 4
```

output

```
Right
```

W. Mathematical Expression

0.25 seconds, 256 MB

Given a mathematical expression. The expression will be one of the following expressions:

$A + B = C$, A - $B = C$ and A * $B = C$

where A, B, C are three numbers, S is the sign between A and B, and Q the '=' sign

Print "Yes" If the expression is **Right** , Otherwise print **the right answer of the expression**.

Input

Only one line containing the expression: A, S, B, Q, C respectively $(0 \le A, B \le 100,\ -10^5 \le C \le 10^5)$ and S can be ('+', '-', '*') without the quotation.

Output

Output either "Yes" (without the quotation) or the right answer depending on the statement.

input
```
5 + 10 = 15
```
output
```
Yes
```

input
```
3 - 1 = 2
```
output
```
Yes
```

input
```
2 * 10 = 19
```
output
```
20
```

X. Two intervals

1 second, 256 megabytes

Given the boundaries of **2** intervals. Print the boundaries of their **intersection**.

Note: **Boundaries** mean the two ends of an interval which are the starting number and the ending number.

Input

Only one line contains two intervals $[l_1, r_1]$, $[l_2, r_2]$ where $(1 \le l_1, l_2, r_1, r_2 \le 10^9)$, $(l_1 \le r_1, l_2 \le r_2)$.

It's guaranteed that $l_1 \le r_1$ and $l_2 \le r_2$.

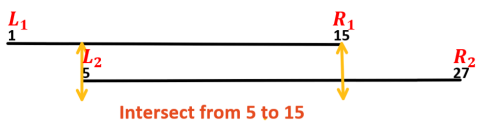
Output

If there is an **intersection** between these **2** intervals print its boundaries , otherwise print **-1**.

input
```
1 15 5 27
```
output
```
5 15
```

input
```
2 5 6 12
```
output
```
-1
```

First Example :



Second Example :

Y. The last 2 digits

1 second, 256 megabytes

Given 4 numbers A, B, C and D. Print the **last 2 digits** from their **Multiplication**.

Input

Only one line containing four numbers A, B, C and D $(2 \le A, B, C, D \le 10^9)$.

Output

Print the **last 2 digits** from their **Multiplication**.

input
```
5 7 2 4
```
output
```
80
```

input
```
3 9 9 9
```
output
```
87
```

First Example :

the Multiplication of 4 numbers is **5 * 7 * 2 * 4 = 280** so the answer will be the last 2 digits which are **80**.

Second Example :

the Multiplication of 4 numbers is **3 * 9 * 9 * 9 = 2187** so the answer will be the last 2 digits which are **87**.

Z. Hard Compare

1 second, 256 megabytes

Given **4** numbers A, B, C and D. If $A^B > C^D$ print "**YES**" otherwise, print "**NO**".

Input

Only one line containing **4** numbers A, B, C and D $(1 \le A, C \le 10^7)$, $(1 \le B, D \le 10^{12})$

Output

Print "**YES**" or "**NO**" according to the problem above.

input
```
3 2 5 4
```
output
```
NO
```

input
```
5 2 4 2
```
output
```
YES
```

input
```
5 2 5 2
```
output
```
NO
```

First Example :

3^2 = 9 and 5^4 = 625 then **9 < 625** so the answer is **NO**.

Second Example :

5^2 = 25 and 4^2 = 16 then **25 > 16** so the answer is **YES**.

Third Example :

5^2 = 25 and 5^2 = 25 then **25 = 25** so the answer is **NO**.